FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2023

TEST DATE: 28-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

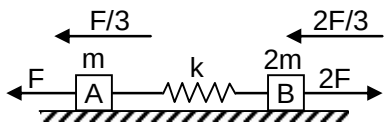
Sol. Centre of mass will be on angle bisector and distance is $\frac{2}{3}R\frac{\sin\frac{\theta_0}{2}}{\frac{\theta_0}{2}}$

x-coordinate = $\left(\frac{2}{3}R\frac{\sin\frac{\theta_0}{2}}{\frac{\theta_0}{2}}\right)\cos\frac{\theta_0}{2}$

y-coordinate = $\left(\frac{2}{3}R\frac{\sin\frac{\theta_0}{2}}{\frac{\theta_0}{2}}\right)\sin\frac{\theta_0}{2}$

2. B

Sol. In frame of centre of mass $\rightarrow a_{cm}=\frac{F}{3m}$



Work energy theorem in centre of mass frame $\rightarrow$

$$\frac{4F}{3}x_1+\frac{4F}{3}x_2=\frac{1}{2}k(x_1+x_2)^2-\frac{1}{2}m(2u)^2-\frac{1}{2}(2m)u^2$$

3. B

Sol. $\Delta H_1 = \frac{Q^2}{2}\left(\frac{1}{C} - \frac{1}{2C}\right)$

$\Delta H_2 = \frac{Q^2}{2}\left(\frac{1}{2C} - \frac{1}{3C}\right)$

.....

.....

$\Delta H_n = \frac{Q^2}{2}\left(\frac{1}{nC} - \frac{1}{(n+1)C}\right)$

$\Delta H = \Delta H_1 + \Delta H_2 + ... = \frac{Q^2}{2C} = \frac{C^2V^2}{2C} = \frac{CV^2}{2}$

$= 2\pi\varepsilon_0 RV^2$

4. D

Sol. Add linear and Angular

EMF $\frac{B\omega L^2}{2} = \frac{BvR}{2}$

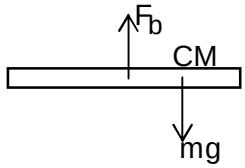
$BvR\sin\omega t = B$

5. A

Sol. Velocity will be sinusoidal function of time.

6. B

Sol. Torque about CM;



$F_b \frac{\ell}{4} = I\alpha$

$\alpha = \frac{\pi r^2 \ell^2 g\rho}{4I}$

7. B

Sol. Upthrust on the complete solid cone $\Rightarrow B = \frac{\pi}{3}(2R)R^2\rho g$

$\Rightarrow F_{Base} - F_{C\,surface} = \frac{2\pi R^2\rho g}{3}$

$\Rightarrow F_1 = F_{C surface} = \left[\rho_0 + \rho g(2R)\right]\pi R^2 - \frac{2}{3}\pi R^2\rho g$ .....(1)

For remaining cone

$F_2 = B.F^{\cdot} + F_1$

$= \left[\frac{\pi}{3}R^2 \times 2R - \frac{\pi}{3}R^3 R\right]\rho g + P_0\left(\pi R^2\right) + \frac{4}{3}\pi R^3\rho g$

$F_1 = \frac{\pi}{3}R^3\rho g\left[5 + \frac{3P_0}{R\rho g}\right]$

8. C

Sol. Let length of semi major axis is a and eccentricity is e.

$$-\frac{GMm}{R_e}+\frac{1}{2}m\left(\sqrt{\frac{GM_e}{R_e}}\right)^2=-\frac{GM_em}{2a}\Rightarrow a=R_e$$

As the separation between ends of semi minor axis and focus is equal to length of semi major axis we can say that launching and landing sites are ends of semi minor axis.

$\Rightarrow R_e\cos45^\circ = ae \Rightarrow e = 1/\sqrt{2}$

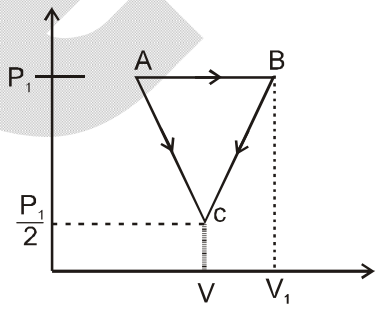
Then maximum separation will be $a + ae = R_e\dfrac{\left(\sqrt{2}+1\right)}{\sqrt{2}}$

9. B

Sol. In the circular motion around the Earth, the centripetal force on the satellite is a gravitational force. Therefore, $v^2 = GM/R$, where M is the mass of the Earth, R is the radius of the orbit of satellite and G is the universal gravitational constant. Therefore, the kinetic energy increases with the decrease in the radius of the orbit. The gravitational potential energy is negative and decreases with the decrease in radius.

10. A

Sol. In equilateral DABC given that $P_A = 2P_C = P_1$



$$\tan 60^0 = \frac{P_1/2}{v_1-v}$$

$$v_1-v=\frac{p_1}{2\sqrt{3}}$$

Work done = area of triangle = $\dfrac{1}{2}2x\left(v_1-v\right)x\dfrac{P_1}{2}=\dfrac{P_1^2}{4\sqrt{3}}$

By ideal gas equation

$$p_1v_1=nRT_1$$

$$p_1=\frac{nRT_1}{V_1}$$

$$w=\frac{\left(nRT_1\right)^2}{4\sqrt{3}v_1^2}=\frac{n^2R^2T_1^2}{4\sqrt{3}v_1^2}$$

N = 2 moles

Work done = $\dfrac{R^3T_1^2}{\sqrt{3}v_1^2}$

11. A

Sol. For circular orbit,

$$\frac{GMm}{r^2}=mw^2r$$ (r = radius of planet's orbit)

$$w^2=\frac{GM}{r^3}=\left(\frac{2\pi}{T}\right)^2 \Rightarrow r=\left(\frac{GMT}{4\pi^2}\right)^{3/2}$$

Energy radiated by the sun per second = $s(4pR^2)q_0^4$

Energy received by the planet persecond = $s(4pR^2)q_4^4\left(\dfrac{\pi R_P^2}{4\pi r^2}\right)$

In thermal equilibrium

$$\sigma.\left(4\pi R^2\right)\theta_0^4\left(\frac{\pi R_P^2}{4\pi r^2}\right)=\sigma.\left(4pR_P^2\right)\theta^4$$

Energy received by planet $\left(\theta=\sqrt{\frac{R}{2R}}\theta_0\right)$ — Energy radiated by planet

12. C

Sol. Let N_o be initial number of active nuclides of ^{238}U

Given $\frac{N(Pb)}{N(U)}=\frac{1}{2} \Rightarrow \frac{N_o-N_oe^{-\lambda t}}{N_oe^{-\lambda t}}=\frac{1}{2} \Rightarrow e^{-\lambda t}-1=\frac{1}{2}$

$$\Rightarrow t=\frac{T_{1/2}\ln(1.5)}{\ln 2}=4.5\times10^9\frac{\ln\left(\frac{3}{2}\right)}{\ln 2}\text{ year}$$

13. A

Sol. $m = m_0 e^{-\lambda t}$

Thrust force on the body is

$$F=u_r\left(-\frac{dm}{dt}\right)=m\left(\frac{dv}{dt}\right)$$

14. B

Sol. Along common normal $\nu_{l/o} = -2\hat{i}$

$\vec{\nu}_{l/o}=\nu_{l/g}-\nu_{o/g}=0$ along C.N.

Hence (B) is correct

15. A

Sol. $S_1P - S_2P = 2\lambda\cos\theta = \lambda$, $\therefore \cos\theta = ½ \Rightarrow \tan\theta = x/D \Rightarrow x = D\tan60^o$

$x = D\sqrt{3}$

16. A

Sol. $r=\frac{R}{2}$

Using parallel axis theorem

$I_x = I_{cm} + mr^2$ twice

17. A

Sol. $11=10\frac{V+V_A}{V}$

$12=10\frac{V+V_B}{V}$

Solve for ratio $\left(\frac{V_B}{V_A}\right)$

18. C

Sol. $PV = nRT$

$P = P_0 - AV^2$

$\Rightarrow (P_0 - AV^2)V = nRT$

$\Rightarrow P_0 - 3AV^2 = nR\left(\frac{dT}{dV}\right)$

For $T_{max}\ \frac{dT}{dV} = 0$

$\Rightarrow V = \sqrt{\frac{P_0}{3A}} \Rightarrow P = P_0 - A\left(\sqrt{\frac{P_0}{3A}}\right)^2$

$\Rightarrow T_{max} = \frac{2P_0}{3nR}\sqrt{\frac{P_0}{3A}}$

$PV = nRT$

$T = \frac{PV}{nR} = \frac{2P_0}{3nR}\sqrt{\frac{P_0}{3A}}$

19. D

Sol. $v_b = 200\hat{i}$

$v_s = \left[-200\cos\theta\,\hat{i} + 200\sin\theta\right]$

$v_{bs} = \left(200 + 200\cos\theta\right)\hat{i} - 200\sin\theta\,\hat{j}$

$+200\sin\theta = \frac{1}{\sqrt{3}}$

$\left(200 + 200\cos\theta\right)$

20. C

Sol. $d\mu = \int dq\left(\frac{\omega}{2\pi}\right)\pi r^2$

SECTION – B

21. 9

Sol. $x_{cm} = \frac{\int dm \times x}{\int dm} = \frac{\int_0^a \left(\sigma \times 2kx^2\right)dx \times x}{\sigma \times \frac{2}{3}ka^3} = \frac{3}{4}a = \frac{9}{12}a$

22. 6

Sol. $Q_0 = C\left(\varepsilon_1 + \varepsilon_2\right) \times 2$

$Q_0 = 72\mu F$

$\frac{Q_0}{12} = \frac{72}{12} = 6\ \mu C$

23. 2

24. 3

Sol. For a small arc subtending angle $d\theta$ at the centre the inward force due to tension in thread is balanced by outward force due to surface tension

$$2T\sin\left(\frac{d\theta}{2}\right) = SRd\theta$$

$$\Rightarrow Td\theta = SRd\theta$$

$$T = SR$$

25. 7

Sol. $$\frac{dQ}{dt} = mc\left(-\frac{dT}{dt}\right)$$

$$E = \sigma AT^4$$

$$mc\left(-\frac{dT}{dt}\right) = \sigma A\left(T^4 - 0^4\right) = \sigma AT^4$$

$$\frac{dT}{T^4} = \frac{-\sigma A}{mc}dt$$

$$\int_{200}^{100} T^{-4}dT = -\frac{\sigma A}{mc}\int_0^t dt$$

$$\left|\frac{T^{-3}}{-3}\right|_{200}^{100} = -\frac{\sigma A}{mc}t$$

$$t = \frac{mc}{3\sigma A}\left[\frac{1}{T^3}\right]_{200}^{100}$$

$$= \frac{\left(\frac{4}{3}\pi r^3\rho\right)c}{3\sigma \times 4\pi r^2}\left[\frac{1}{100^3} - \frac{1}{200^3}\right]$$

$$= \frac{7}{72}\frac{r\rho c}{\sigma} \times 10^{-6}\ \text{sec.}$$

26. 4

Sol. $$T = \frac{2v_0\sin 37^\circ}{g\cos 37^\circ} = \frac{3}{2}\text{sec}$$

$$V_0\cos 37^\circ \times \frac{3}{2} + \frac{1}{2}g\sin 37^\circ \times \left(\frac{3}{2}\right)^2 = 6 + V \times \frac{3}{2} + \frac{1}{2}g\sin 37^\circ\left(\frac{3}{2}\right)^2$$

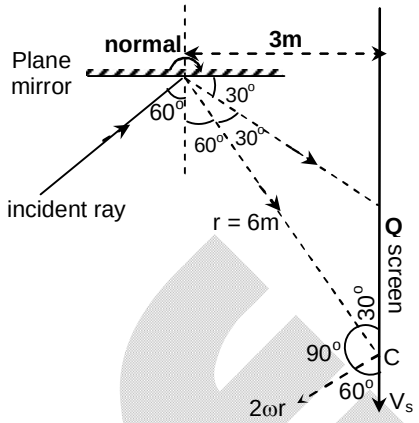
$$V = 4$$

27. 9

Sol. $$\sqrt{f} = a(Z-1)$$

28. 2

Sol. When incident ray is fixed the angular velocity of reflected ray becomes twice in the same sense as that of reflected ray. After t = 15 sec, the mirror rotates 15° then reflected ray rotates 30° in the same sense.



In ΔABC, $\cos 60° = \dfrac{3}{r}$

Where $\omega = \dfrac{\pi}{180}\text{rad/sec}.\ r = 6\text{m}$

Then, there are two components of velocity of spot on the screen.

(a) radial component which increase the length of r i.e.,

$V_s \sin 60° = \dfrac{dr}{dt}$

(b) perpendicular component

$V_s \cos 60° \qquad \Rightarrow V_s \times \dfrac{1}{2} = 2 \times \dfrac{\pi}{180} \times 6 \Rightarrow V_s = \left(\dfrac{2\pi}{15}\text{m/s}\right)$

29. 8

Sol. The stresses are zero at the free end and maximum at the axis. Therefore, the rod will ruptures at the middle.

Let us consider the element of rod at a distance x from the axis, the mass of element

$dm = \rho A dx$

Applying Newton's second law,

$F - (F + dF) = (\rho A dx)\omega^2 x$

Where ω is the rotation speed.

$$F = -\int dF = -\rho A\omega^2 \frac{x^2}{2} + C$$

Rupture of rod will occur when $f = \sigma$

$$\therefore \frac{\rho\omega^2 l^2}{8} = \sigma$$

$$\Rightarrow \omega = \sqrt{\frac{8\sigma}{\rho l^2}} \Rightarrow n = \frac{1}{2\pi}\sqrt{\frac{8\sigma}{\rho l^2}}$$

30. 4

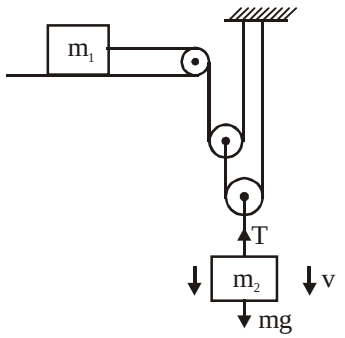
Sol. Applying work energy theorem

$K_f - K_i = w$

$\dfrac{1}{2}m_2v^2 + \dfrac{1}{2}m_1(4v)^2$

$= m_2g \times y - \mu m_1 g(4y)$

v = 4 m/s.



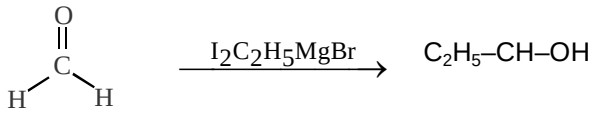
Chemistry

PART – B

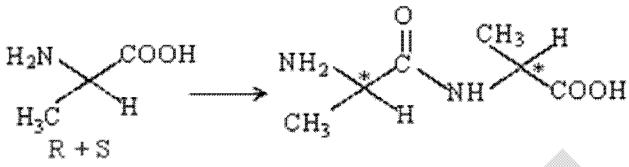
SECTION – A

31. D
Sol. Pentaammine chloride cobalt (III) chloride
$[Co(NH_3)_5Cl]Cl_2$ $[Co(NH_3)5Cl^-]^{+2} + 2Cl^-$
Gives 3 ions in aqueous solution.

32. C
Sol.



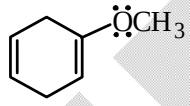
33. D
Sol.



34. B
Sol. Charge on 1 milimole M^{n+} ions = 193 cb

$$= \frac{n \times 96500}{1000}$$

$$n = \frac{193 \times 1000}{96500} = 2$$

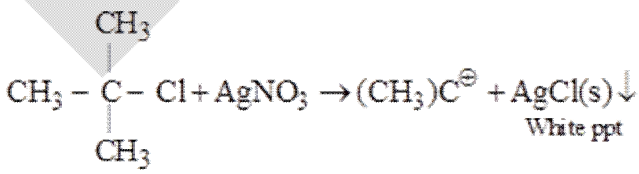
35. D
Sol. In  the lone pair of oxygen

get delocalised on the π bond located on the next carbon.

36. C
Sol. Quantum number set n = 2, ℓ = 0, m = –1 it is not possible (not valid).

37. B
Sol. neq. KMnO4
= neq.$[FeC_2O_4 + Fe_2(C_2O_4)_3 + FeSO_4]$
or $n \times 5 = 1 \times 3 + 1 \times 6 + 1 \times 1$
$\therefore$ n = 2

38. A
Sol.



Reason : Due to most stable carbocation formation tert-butyl chloride given the ppt immediately.

39. C

Sol. $\Delta Tb = Kb \times m$

$$\therefore \quad \frac{\Delta T_{b\,(A)}}{\Delta T_{b\,(B)}} = \frac{K_{b\,(A)}}{K_{b\,(B)}} \text{ as } m_A = m_B$$

$$\therefore \quad \frac{\Delta T_{b\,(A)}}{\Delta T_{b\,(B)}} = \frac{1}{5}$$

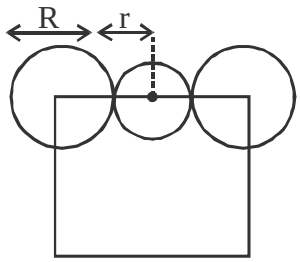
40. D

Sol. Colligative properties of colloidal solution are smaller than true solution.

41. C

Sol. $a = 2\,(R + r)$

$$\frac{a}{2} = (R + r) \ldots(1)$$



$$a\sqrt{3} = 4R \ldots(2)$$

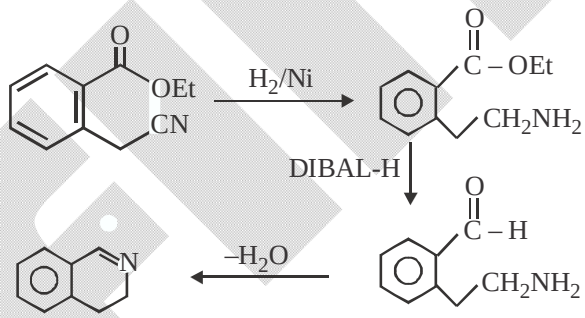
Using (1) & (2)

$$\frac{a}{2} = \frac{a\sqrt{3}}{4} = r \quad ; \quad a\left(\frac{2-\sqrt{3}}{4}\right) = r$$

$$r = 0.067\,a$$

42. B

Sol.



43. B

Sol. A complex having strong field ligand has tendency to absorb light of highest energy. Among the three complexes $[Co(NH_3)_6]^{+3}$ will absorb radiation of highest energy and least wavelength. $[Co(NH_3)_5H_2O]^{3+}$ has field weaker than the above compound and therefore absorb radiation of lesser energy and more wavelength.

$[CoCl(NH_3)_5]^{2+}$ has the weakest field and therefore will absorb light of least energy and highest wavelength.

Strength of ligand $NH_3 > H_2O > Cl$.

44. B
Sol. Reagent based

45. D
Sol. Nitrogen oxides and hydrocarbons (unburnt fuel) are primary pollutant that leads to photochemical smog.

46. C
Sol. NaH is an example of ionic hydride which is also known as saline hydride.

47. B
Sol. $H_2(g) + I_2(g) \rightarrow 2HI\ (g)$
Apply Arrhenius equation

$$\log\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left(\frac{1}{600} - \frac{1}{800}\right)$$

$$\log\frac{1}{2.5\times10^{-4}} = \frac{E_a}{2.303\times8.31}\left(\frac{200}{600\times800}\right)$$

$\therefore$ $E_a \approx 166$ kJ/mol

48. C
Sol. Polarisation power $\propto$ + ve charge
$\propto$ 1 / Size of ion

49. C
Sol.

$$(CH_3)_2CHBr \xrightarrow[(ii)\ CuI]{(i)\ Li} [(CH_3)_2CH]_2CuLi$$

$$\xrightarrow{(CH_3)_2CHCH_2Br} (CH_3)_2CH{-}CH_2{-}CH(CH_3)_2$$

50. B
Sol. $MnO_4^- + Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + Fe^{3+}$
0.1M 1.72M
0.01 M 1.0M 0.09 M

$$E^0_{RP\,(MnO_4^-/Mn^{2+})} > E^0_{RP\,(H^+/H_2)}$$

Permangnate electrode behave as cathode
$E°_{cell} = 1.51$ V
Using Nernsts equation : (n = 5)

$$E^0_{MnO_4^-/Mn^{2+}} = 1.51 + \frac{0.059}{5}\log\frac{[MnO_4^-][H^+]^8}{[Mn^{2+}]}$$

= 1.51 – 0.011 = 1.499 ≈1.5V

SECTION – B

51. 9

Sol. Initial moles of $H_2O_2 = 2 \times 2 = 4$

Moles of H_2O_2 left after 20 min.

$= 4 \times \frac{1}{2^2} = 1$ (2 half lives)

Moles of H_2O_2 decomposed 20 min.

$= 4 - 1 = 3$

$H_2O_2 \rightarrow H_2O + \frac{1}{2} O_2$

Moles of O_2 formed during 20 min. = 3/2

Volume change = $\frac{\frac{3}{2} \times R \times T}{P}$

$q = 0$; $\Delta U = W$

$W = -P_{ext} (V_2 - V_1) = - P_{ext} \Delta V$

$= - \frac{3}{2} \times 2 \times 300 = - 900$ Cal.

52. 7

Sol. (i) Planar molecules :

XeF_2, ClF_3, H_2O, $[XeF_5]^-$, I_3^-, BCl_3, XeF_4.

(ii) SF_4 – See- Saw shape

PCl_5 – Trigonal bipyramidal

SF_6 – Square bipyramidal

IF_7 – Pentagonal bipyramidal

53. 3

Sol. $(P) \xleftarrow[(1)]{CF_3CH_2O^{\Theta}} \boxed{CH_3 - CH_2 - Br} \xrightarrow[(2)]{CH_3CH_2O^{\Theta}}$

In reaction 2nd Nucleophile is stronger.

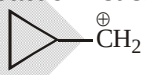
(Q) (Alkene) $\xleftarrow[(CH_3)_3CBr/\Delta]{(1)} \boxed{C_2H_5O^{\Theta}}$

$\xrightarrow[(CH_3)_2CHCH_2Br/\Delta]{(2)}$ (Alkene)

In reaction 1st 3° alkyl halide

1 > 2

(R) In reaction 1st stable carbocation is formed



(Cyclopropyl methyl C^+)

1 > 2

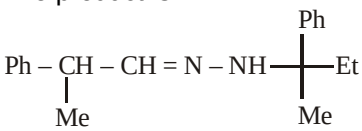
(S) In reaction 1st reagent in HI.

HI is more ionised, gives sufficient concentration of H^+ ion and I^+ is a good Nucleophile also.

> 2

54. 4

Sol. The product is



I = (+, +) (E)
II = (+, +) (Z)
III = (–, +) (E)
IV = (–, +) (Z)

55. 5

Sol. $Na_2SO_3 \xrightarrow{AgNO_3} Ag_2SO_3 \leftarrow$white ppt.

$NaCl \xrightarrow{AgNO_3} AgCl \leftarrow$white ppt.

$Na_2C_2O_4 \xrightarrow{AgNO_3} Ag_2C_2O_4 \leftarrow$white ppt.

$Na_2HPO_4 \xrightarrow{AgNO_3} Ag_3PO_4 \leftarrow$yellow ppt.

$Na_2CrO_4 \xrightarrow{AgNO_3} Ag_2CrO_4 \leftarrow$red ppt.

$NaNO_2 \xrightarrow{AgNO_3} AgNO_2 \leftarrow$white ppt.

$CH_3COONa \xrightarrow{AgNO_3} CH_3COOAg \leftarrow$white ppt.

56. 2

Sol. $PbO_2 + 4KI \longrightarrow PbI_2 + I_2 + 2K_2O$

$N_2H_4 + 2I_2 \longrightarrow N_2 + 4HI$

moles of N_2 liberated $= \frac{1.12}{22.4} = 5 \times 10^{-2}$ mole

sol mole of I_2 reacted $=5\times10^{-2} \times 2 = 10^{-1}$ mole

$PbO_2 + 2H_2O \longrightarrow Pb(OH)_4$

$Pb(OH)_4 + 2NaOH \longrightarrow Na_2[Pb(OH)_6]$

Na_2PbO_2 mole of NaOH required = $10^{-1}\times 2$

$V \times \frac{1}{10} \times 10^3 = 2 \times 10^{-1}$

$V = \frac{2}{10^{-3}} = 2000$ mili litre = 2 litre

57. 4

Sol. The Be^{3+} ion has one electron in the 1s orbital (n = 1) in its ground state. Since beryllium has four protons in its nucleus (Z = 4), the energy of the electron in the 1s orbital of Be^{3+} can be calculated.

$$E = -\left(\frac{Z^2}{n^2}\right)(1310)\ kJ/mole$$

$$= -\left(\frac{4^2}{1^2}\right)(1310)\ \text{kJ/mole}$$

$= 2.10 \times 10^4$ kJ/mol

58. 4

Sol. U_{rms} at temp $T_1 = \sqrt{\frac{3RT_1}{M}}$

U_{rms} at temp $T_2 = \sqrt{\frac{3RT_2}{M}}$

$\therefore \quad \frac{U_{rms}\ \text{at temp.}\ T_2}{U_{rms}\ \text{at temp.}\ T_1} = \sqrt{\frac{T_2}{T_1}}$

or $\quad \frac{10\times10^4}{5\times10^4} = \sqrt{\frac{T_2}{T_1}} \quad \text{or} \quad \sqrt{\frac{T_2}{T_1}} = \frac{4}{1}$

$\therefore$ Temperature is to be increased 4 times.

59. 4

Sol. We can write, $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

Initial pressure : 0.2 atm 0 0

Equilibrium partial pressure :

0.04 atm $\quad \frac{(0.2-0.04)}{2} \quad \frac{(0.2-0.04)}{2}$

$= 0.08 \quad = 0.08$

Then, $K_p = \frac{[H_2]\,[I_2]}{[HI]^2} = \frac{0.08\times0.08}{(0.04)^2} = 4.0$

60. 400

Sol. Equivalents of Na_2CO_3 used

$= (100 - 60) \times 10^{-3} \times \frac{1}{50}$

Equivalent of Na_2CO_3 used = equivalent of Ca^{2+} or Mg^{2+} present in 100 ml of hard water

= equivalent of Ca^{2+} or Mg^{2+}
present in 100 ml of hard water.

The amount of $CaCO_3$ in 10^3 g of hard water = $\frac{40\times10^{-3}\times10\times1\times50}{50} = 400 \times 10^{-3}$ g

The amount of $CaCO_3$ in 10^6 g of hard water
= 400 g.
Permanent hardness of water = 400 ppm.

Mathematics

PART – C

SECTION – A

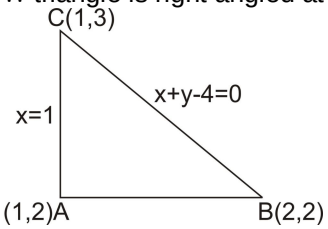
61. B

Sol. $(x+y-4)(xy-2x-y+2)=0$

$\Rightarrow (x+y-4)(x-1)(y-2)=0$

$\therefore$ sides of the triangle are $x+y-4=0, x=1, y=2$

$\therefore$ triangle is right angled at (1, 2).



Hypotenuse is x + y = 4, whose ends are (1, 3), (2, 2).

$\therefore$ circumcentre is $\left(\frac{2+1}{2},\frac{2+3}{2}\right)$ i.e. $\left(\frac{3}{2},\frac{5}{2}\right)$.

circumradius $=\frac{1}{2}\sqrt{(1-2)^2+(3-2)^2}=\frac{1}{\sqrt{2}}$

$\therefore$ circumcircle is $\left(x-\frac{3}{2}\right)^2+\left(y-\frac{5}{2}\right)^2=\frac{1}{2}$

$\Rightarrow x^2+y^2-3x-5y+\frac{9}{4}+\frac{25}{4}-\frac{1}{2}=0$

$\Rightarrow \quad x^2+y^2-3x-5y+8=0$

62. C

Sol. Clearly (c) is the right choice.

$\therefore$ length of the $\perp$ from (4, 5) upon 3x + 4y – 7 = 0 is

$$\frac{12+20-7}{\sqrt{9+16}}=\frac{25}{5}=5$$

Also for (-2, -3).

63. A

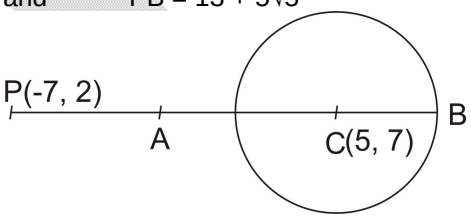
Sol. The centre C of the circle is (5, 7)

and radius = $\sqrt{5^2+7^2+51}=5\sqrt{5}$

PC = $\sqrt{(5+7)^2+(7-2)^2}=13$

$\therefore$ PA = 13 - 5√5

and PB = 13 + 5√5



$\therefore \quad p = 13 + 5\sqrt{5}$

$q = 13 - 5\sqrt{5}$

G.M. of $p,q = \sqrt{pq} = \sqrt{169 - 125} = \sqrt{44} = 2\sqrt{11}.$

64. A

Sol. Equation of any line through the given pt. (13, 31) is $y - 31 = m(x - 13)$.

It meets the x – axis at Q $\left(13 - \frac{31}{m}, 0\right)$ and y – axis at S $(0, 31 - 13m)$

$\therefore$ the co – ordinates of R are (h, k), where $h = 13 - \frac{31}{m}, k = 31 - 3m$

$\therefore (h - 13)(k - 31) = 31 \times 13$

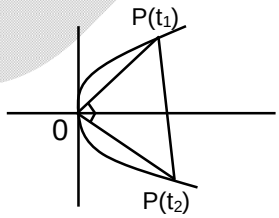
$\Rightarrow hk = 31h + 13k \quad \Rightarrow \frac{13}{h} + \frac{31}{k} = 1$

Hence R satisfies $\frac{13}{x} + \frac{31}{y} = 1.$

65. D

Sol. $t_2 = -t_1 - \frac{2}{t_1}$ slope of $OP = \frac{2}{t_1} = M$

slope of $OQ = \frac{2}{t_2} = N$



$\frac{2}{t_1} \times \frac{2}{t_2} = -1$

So, $t_1 = \pm\sqrt{2}$

66. A

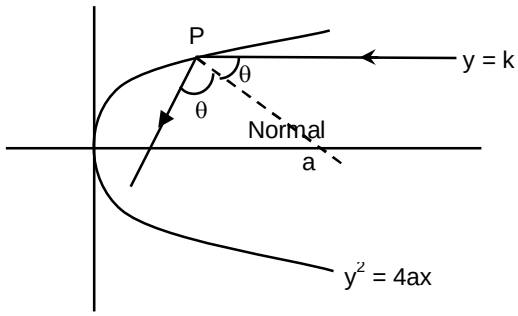
Sol. The line $y = k$ intersects parabola $y^2 = 4ax$ at point $P\left(\frac{k^2}{4a}, k\right)$

Slope of normal at P is $m = \frac{-k}{2a}$

Slope of incident ray is O.

At slope of reflected ray is m,

$$\frac{-k}{2a} = \frac{m_1 - \left(\frac{-k}{2a}\right)}{1 + m_1\left(-\frac{k}{2a}\right)} \Rightarrow m_1 = \frac{4ak}{k^2 - 4a^2}$$



Equation of reflected ray is $(k^2 - 4a^2)y = 4ak(x - a)$.

67. B

Sol. $P(t_1) \qquad Q(t_2) \qquad T(at_1t_2, a(t_1 + t_2))$

Let equation of tangent be x = 0

Hence

$P_1 = at_1^2 \qquad P_2 = at_1t_2 \qquad P_3 = at_2^2$

$P_1P_3 = P_2^2.$

68. B

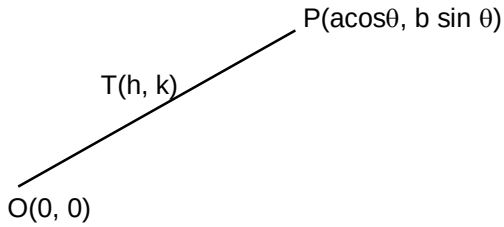
Sol.

$2h = a\cos\theta \quad \Rightarrow \quad \cos^2\theta = \left(\frac{2h}{a}\right)^2$

$2k = b\sin\theta \quad \Rightarrow \quad \sin^2\theta = \left(\frac{2k}{b}\right)^2$

$\frac{4h^2}{a^2} + \frac{4k^2}{b^2} = 1$

$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{4}}$



69. C

Sol. A normal at P(a cos θ, b sin θ) is

a x secθ – by cosec θ = $a^2 - b^2$

$T\left(\frac{a^2-b^2}{a}\cos\theta, 0\right)$ $\qquad$ $G\left(0, \frac{a^2-b^2}{a}\sin\theta\right)$

$PT^2 = \frac{b^2}{a^2}(a^2\sin^2\theta + b^2\cos^2\theta)$

$PG^2 = \frac{a^2}{b^2}(a^2\sin^2\theta + b^2\cos^2\theta)$

$PT : PG = \frac{b^2}{a^2} = 1 - e^2$

70. A

Sol. $\underset{x\to-\infty}{Lim}\left(x^{5c}\left(1+\frac{7}{x}+\frac{2}{x^5}\right)^c - x\right) = \underset{x\to-\infty}{Lim}\, x\left(x^{5c-1}\left(1+\frac{7}{x}+\frac{2}{x^5}\right)^c - 1\right)$

This will be of the form ∞ × 0 only if

5c – 1 = 0 c = 1/ 5 substituting c = 1/5 , l becomes

Hence $l = \lim_{x\to-\infty} x\left[(1+x)^{1/5} - 1\right]$ *where* $X = \frac{7}{x} + \frac{2}{x^5}$

$= \lim_{x\to-\infty} x\left[1+\frac{x}{5}+....-1\right] \lim_{x\to-\infty} x\left(\frac{7}{x}+\frac{2}{x^5}\right).\frac{1}{5} = \frac{7}{5}$

Hence $c = \frac{1}{5}$ *and* $l = \frac{7}{5}$

71. C

Sol. [Hint: $A = \frac{1}{3}, B = -1, C = \frac{2}{3}; D = \frac{13}{9}$]

72. D

Sol. [Sol. since 0 < sin x < 1 and 1 + x > 1 in (0, π/2)

hence $I_3 > I_2 > I_1$

⇒ A and B are correct ⇒ (D)]

73. D

Sol. Let f(x) = 2 tan x + (2a + 1) ln |sec x| + (a – 2)x

⇒ f'(x) = 2 $\sec^2$x + (2a + 1) tan x + (a – 2) = 2 $\tan^2$ x + (2a + 1) tan x + a

For f(x) to be increasing, $f'(x) \geq 0 \ \forall \ x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow \quad 2\tan^2 x + (2a+1)\tan x + a \geq 0 \forall x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow \quad (1+2\tan x)(\tan x + a) \geq 0 \forall x \in \left(0, \frac{\pi}{2}\right)$

Hence $a \geq 0$. Ans

74. D

Sol. Find maximum of $\left(\frac{7-5(x^2+3)}{2(x^2+2)}\right)$

75. A

Sol. Since given system of equation has a non – trivial solution.

$$\therefore \begin{vmatrix} p+a & b & c \\ a & q+b & c \\ a & b & r+c \end{vmatrix} = 0$$

Operate $R_1 - R_2$ and $R_2 - R_3$, we get $\begin{vmatrix} p & -q & 0 \\ 0 & q & -r \\ a & b & r+c \end{vmatrix} = 0$

$$\Rightarrow p\begin{vmatrix} q & -r \\ b & r+c \end{vmatrix} + a\begin{vmatrix} -q & 0 \\ q & -r \end{vmatrix} = 0 \qquad \Rightarrow p(qr+qc+br) + a(qr) = 0$$

$$\Rightarrow 1 + \frac{c}{r} + \frac{b}{q} + \frac{a}{p} = 0 \qquad \Rightarrow \frac{a}{p} + \frac{b}{q} + \frac{c}{r} = -1.$$

76. C

Sol. Clearly $ab - h^2 \neq 0$ [$\because$ first two equations are solvable for x, y]

Third equation can be written as

$x(ax + hy + g) + y(hx + by + f) + gx + fy + c + t = 0$

Since any solution of first two must be sol. of the third equation.

$\therefore$ given system of equations can be written as

$ax + hy + g = 0$

$hx + by + f = 0$

$gx + fy + c + t = 0$

Since the above system of equations are consistent

$$\therefore \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c+t \end{vmatrix} = 0 \qquad \Rightarrow \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} + \begin{vmatrix} a & h & 0 \\ h & b & 0 \\ g & f & t \end{vmatrix} = 0$$

$$\Rightarrow \left(abc + 2fgh - af^2 - bg^2 - ch^2\right) + t\left(ab - h^2\right) = 0$$

$$\Rightarrow t = \frac{abc + 2fgh - af^2 - bg^2 - ch^2}{(h^2 - ab)}$$

77. A

Sol. $\overrightarrow{OR} = \frac{2\vec{q} + 3\vec{p}}{2+3} = \frac{2\vec{q} + 3\vec{p}}{5}$

$\overrightarrow{OS} = \frac{2\vec{q} - 3\vec{p}}{2-3} = \frac{2\vec{q} - 3\vec{p}}{-1}$

Since $\overrightarrow{OR} \perp \overrightarrow{OS} \therefore \overrightarrow{OR}.\overrightarrow{OS} = 0$

$\therefore \frac{2\vec{q}+3\vec{p}}{5}.\frac{2\vec{q}-3\vec{p}}{-1} = 0$

$\Rightarrow \quad \frac{4|\vec{q}|^2 - 9|\vec{p}|^2}{-5} = 0 \quad \Rightarrow 4q^2 - 9p^2 = 0 \Rightarrow \quad 9p^2 = 4q^2$

78. A

Sol. Now $\vec{d}.(\vec{a}+\vec{b}+\vec{c}) = \vec{d}.\vec{a} + \vec{d}.\vec{b} + \vec{d}.\vec{c}$

$= \mu(\vec{b}\times\vec{c}.\vec{a}) + \gamma(\vec{c}\times\vec{a}.\vec{b}) + \lambda(\vec{a}\times\vec{b}.\vec{c})$

$= \lambda + \mu + \gamma \qquad [\because (\vec{a},\vec{b},\vec{c}) = 1]$

Hence $\lambda + \mu + v = \vec{d}.(\vec{a}+\vec{b}+\vec{c})$

79. C

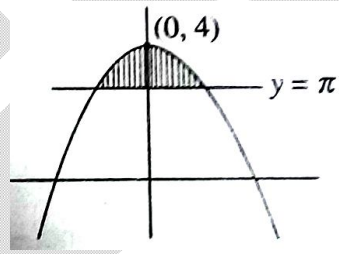
Sol. $0 \le \sin^2 x \le 1 \Rightarrow [-\sin^2 x] = 0$ or -1

But $\sec^{-1}(0)$ is not defined

$\Rightarrow y = \sec^{-1}[-\sin^2 x] = \sec^{-1}(-1) = \pi$

Solving with $y = \frac{16-x^2}{4}, 16 - x^2 = 4\pi$

$\therefore \qquad x = \pm 2\sqrt{4-\pi}$



Required area = $\int_{-2\sqrt{4-\pi}}^{2\sqrt{4-\pi}} \left(\frac{16-x^2}{4} - \pi\right) dx = \frac{8}{3}(4-\pi)^{3/2}$

80. A

Sol. The total no. of ways of predicting the result is 3^{10}. The no. of ways in which exactly K results are correct is ${}^{10}C_K \cdot 2^{10-K}$

$\therefore$ the no. of favourable ways is

${}^{10}C_5 \cdot 2^5 + {}^{10}C_6 \cdot 2^4 + {}^{10}C_7 \cdot 2^3 + {}^{10}C_8 \cdot 2^2 + {}^{10}C_9 \cdot 2 + {}^{10}C_{10} = 12585.$

Hence reqd. prob. $= \frac{12585}{3^{10}}$

SECTION – B

81. 3

Sol. sinx + cosx + tanx + cotx + secx + cosecx = 7

$\Rightarrow (\sin x + \cos x) + \left(\frac{1}{\sin x \cos x}\right) + \frac{(\sin x + \cos x)}{\sin x.\cos x} = 7$

$\Rightarrow (\sin x + \cos x)\left(1 + \frac{2}{\sin 2x}\right) = \left(7 - \frac{2}{\sin 2x}\right)$

$\Rightarrow (1 + \sin 2x)\left(1 + \frac{4}{\sin^2 2x} + \frac{4}{\sin 2x}\right) = 49 + \frac{4}{\sin^2 2x} - \frac{28}{\sin 2x}$ (squaring both sides)

$\Rightarrow \sin^3 2x - 44 \sin^2 2x + 36 \sin 2x = 0$

$\Rightarrow \sin 2x = 22 - 8\sqrt{7}$.

82. 4

Sol. $\tan^4x - 2\tan^3 x - \tan^2x + 2\tan x + 1$
$\tan^4x + \tan^2x - 2\tan^3x + 2\tan x - 2\tan^2x + 1$
$= (\tan^2 - \tan x)^2 + 2(\tan x - \tan^2x) + 1 = 4.$

83. 2

Sol. Use angle between asymptotes as $\theta = \tan^{-1}\dfrac{2ab}{b^2 - a^2}$ & $\text{Latusrectum} = \dfrac{2b^2}{a}$

84. 5

Sol. It is point of contact of tangent

85. 11

Sol. $x \in (-\infty, 0] \cup [\log_6 5, \infty)$

86. 6

Sol. S.D (and variance) of a data is not changed when each observation is increased (or decreased) by the same constant.

87. 0

Sol. $[u = \int_0^1 \frac{ln(x+1)}{x^2+1}dx \qquad put\ x = \tan\theta$

$= \int_0^{\pi/4} \ln(1+\tan\theta)d\theta == \int_0^{\pi/4} \ln\left(1+\frac{1-\tan\theta}{1+\tan\theta}\right)d\theta == \int_0^{\pi/4} \ln\left(\frac{2}{1+\tan\theta}\right)d\theta = \frac{\pi}{4}\ln 2 - u$

$\therefore \qquad u = \frac{\pi}{8}\ln 2 \qquad \Rightarrow 4u = \frac{\pi}{2}\ln 2 \quad ....(1)$

Again $v = \int_0^{\pi/2} \ln(\sin 2x)dx \qquad (put\ 2x = t); v = \frac{1}{2}\int_0^{\pi} \ln(\sin t)dt = \int_0^{\pi/2} \ln(\sin t)dx$

$v = -\frac{\pi}{2}\ln 2 \qquad ...(2)$

$(1)+(2) \Rightarrow \qquad 4u + v = 0 \qquad \Rightarrow (C)]$

88. 2

Sol. Using Graph number of points of non differentiability = 2 Ans.

89. 5

Sol. $f'(x) = 2x - \dfrac{a}{x^2}$

$f'(2) = 4 - \frac{a}{4} = 0 \Rightarrow a = 16 \Rightarrow a_1 = 16$

Also, $f''(x) = 2 + \dfrac{2a}{x^3}$

$\therefore \ f''(1) = 2 + 2a = 0 \Rightarrow a = -1 \Rightarrow a_2 = -1$

Hence $\left(\dfrac{a_1 + a_2}{3}\right) = \dfrac{15}{3} = 5$ Ans.

90. 11

Sol. The last digit of the product will be 1, 2, 3, 4, 6, 7, 8 or 9 iff each of the n positive integer ends in any of these digits. Now the probability of an integer ending in 1, 2, 3, 4, 6, 7, 8 or 9 is $\frac{8}{10}=\frac{4}{5}$

$\therefore$ The prob. that the last digit of the product of n integers is 1, 2, 3, 4, 6, 7, 8 or 9 = $\left(\frac{4}{5}\right)^n$.

Next, the last digit of the product will be 1, 3, 7 or 9 iff each of the n + ve integers ends in 1, 3, 7 or 9 is $\frac{4}{10}=\frac{2}{5}$

$\therefore$ The prob. for the product of n + ve integer to end in 1, 3, 7 or 9 is $\left(\frac{2}{5}\right)^n$

Hence the prob. of the reqd. event = $\left(\frac{4}{5}\right)^n-\left(\frac{2}{5}\right)^n=\frac{4^n-2^n}{5^n}$